

## ARANYAPRATHET, Thailand

WMO: 484620

Lat: 13.6886N

Lon: 102.5042E

Elev: 161

StdP: 14.61

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 64.5       | 67.1 | 52.7                          | 59.3 | 71.8 | 55.4 | 65.8 | 73.2 | 13.9                  | 79.7 | 12.2 | 79.5 | 1.1                   | 30   | 0.333 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 17.9                   | 100.2           | 79.8 | 98.4 | 79.4 | 96.7 | 79.2 | 82.5                | 93.8 | 81.8 | 92.7 | 81.3 | 91.8 | 3.3                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 80.0                            | 156.4 | 86.8 | 79.2 | 152.5 | 86.0 | 78.8 | 150.5 | 85.6 | 46.5          | 94.3 | 45.7 | 93.2 | 45.2 | 92.0 | 90.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%  | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |     | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 10.7              | 9.1  | 7.6 | 60.1                       | 102.7 | 2.9                | 1.8 | 58.0                                               | 104.0 | 56.3       | 105.1 | 54.6       | 106.1 | 52.5       | 107.4 |
|                   |      |     | 55.2                       | 84.8  | 3.0                | 1.9 | 53.0                                               | 86.2  | 51.3       | 87.4  | 49.6       | 88.4  | 47.5       | 89.9  |
|                   |      |     | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |     | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 84.0  | 80.4  | 83.9  | 86.8  | 87.9  | 86.9  | 85.6  | 84.3  | 84.0  | 83.5  | 83.3  | 81.8  | 79.6 | (6)  |
| (7)  |                                                                                  | DBStd       | 3.79  | 3.84  | 3.71  | 3.12  | 2.94  | 2.70  | 2.26  | 2.17  | 1.96  | 1.98  | 2.08  | 3.05  | 3.92 | (7)  |
| (8)  |                                                                                  | HDD50       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD65       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD50       | 12411 | 943   | 950   | 1140  | 1137  | 1143  | 1067  | 1063  | 1055  | 1006  | 1034  | 955   | 918  | (10) |
| (11) |                                                                                  | CDD65       | 6936  | 478   | 530   | 675   | 687   | 678   | 617   | 598   | 590   | 556   | 569   | 505   | 453  | (11) |
| (12) |                                                                                  | CDH74       | 79703 | 5093  | 6299  | 8597  | 8905  | 8369  | 7297  | 6687  | 6489  | 5893  | 6157  | 5358  | 4559 | (12) |
| (13) |                                                                                  | CDH80       | 36227 | 2300  | 3161  | 4583  | 4850  | 4129  | 3264  | 2689  | 2544  | 2180  | 2399  | 2223  | 1905 | (13) |
| (14) | Wind                                                                             | WSAvg       | 1.7   | 1.7   | 1.7   | 1.8   | 1.7   | 1.7   | 1.8   | 1.8   | 1.4   | 1.4   | 2.0   | 2.3   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 52.3  | 0.3   | 1.0   | 2.1   | 3.6   | 5.7   | 6.5   | 6.9   | 7.8   | 9.6   | 7.1   | 1.5   | 0.2  | (15) |
| (16) |                                                                                  | PrecMax     | 71.5  | 1.2   | 4.2   | 5.8   | 8.5   | 12.8  | 12.4  | 13.7  | 13.9  | 18.9  | 12.8  | 4.4   | 1.3  | (16) |
| (17) |                                                                                  | PrecMin     | 37.2  | 0.0   | 0.0   | 0.2   | 0.7   | 1.3   | 3.0   | 2.9   | 3.0   | 3.0   | 3.1   | 0.0   | 0.0  | (17) |
| (18) |                                                                                  | PrecStd     | 7.2   | 0.3   | 1.1   | 1.9   | 2.1   | 2.6   | 2.4   | 2.4   | 2.7   | 3.2   | 2.8   | 1.4   | 0.3  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 95.4  | 98.9  | 101.9 | 103.4 | 101.3 | 98.1  | 95.7  | 94.6  | 94.2  | 94.0  | 94.8  | 94.7 | (19) |
| (20) |                                                                                  |             | MCWB  | 76.4  | 77.4  | 78.5  | 80.4  | 81.2  | 80.7  | 80.5  | 80.9  | 80.8  | 79.8  | 78.7  | 77.3 | (20) |
| (21) |                                                                                  | 2%          | DB    | 93.7  | 96.9  | 99.6  | 100.7 | 98.4  | 95.6  | 93.6  | 93.0  | 92.5  | 92.5  | 93.1  | 92.7 | (21) |
| (22) |                                                                                  |             | MCWB  | 75.7  | 77.2  | 78.3  | 80.4  | 81.0  | 80.6  | 80.3  | 80.5  | 80.4  | 79.6  | 78.3  | 76.2 | (22) |
| (23) |                                                                                  | 5%          | DB    | 92.2  | 95.3  | 97.8  | 98.6  | 96.1  | 93.7  | 92.1  | 91.5  | 90.9  | 91.2  | 91.4  | 90.9 | (23) |
| (24) |                                                                                  |             | MCWB  | 75.0  | 76.8  | 78.2  | 80.3  | 80.9  | 80.4  | 79.9  | 80.1  | 80.0  | 79.4  | 77.7  | 75.3 | (24) |
| (25) |                                                                                  | 10%         | DB    | 90.3  | 93.4  | 95.8  | 96.3  | 93.9  | 92.0  | 90.4  | 90.0  | 89.3  | 89.7  | 89.7  | 89.1 | (25) |
| (26) |                                                                                  |             | MCWB  | 74.2  | 76.4  | 78.1  | 79.9  | 80.6  | 80.2  | 79.5  | 79.7  | 79.6  | 79.0  | 77.1  | 74.6 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 80.1  | 80.8  | 81.4  | 82.9  | 83.5  | 82.8  | 82.1  | 83.1  | 82.2  | 81.9  | 80.9  | 79.8 | (27) |
| (28) |                                                                                  |             | MCDWB | 91.4  | 94.5  | 95.4  | 97.4  | 96.0  | 92.6  | 92.0  | 91.4  | 91.0  | 90.3  | 90.8  | 90.3 | (28) |
| (29) |                                                                                  | 2%          | WB    | 78.2  | 79.4  | 80.4  | 81.9  | 82.4  | 81.8  | 81.1  | 81.6  | 81.1  | 80.9  | 80.0  | 78.2 | (29) |
| (30) |                                                                                  |             | MCDWB | 89.2  | 92.6  | 94.2  | 96.0  | 94.3  | 91.6  | 90.6  | 90.0  | 89.7  | 89.4  | 89.4  | 88.8 | (30) |
| (31) |                                                                                  | 5%          | WB    | 77.0  | 78.6  | 79.8  | 81.2  | 81.7  | 81.2  | 80.5  | 80.8  | 80.6  | 80.3  | 79.2  | 77.2 | (31) |
| (32) |                                                                                  |             | MCDWB | 87.5  | 91.2  | 93.1  | 94.6  | 92.8  | 90.7  | 89.7  | 89.1  | 88.9  | 88.6  | 88.1  | 87.1 | (32) |
| (33) |                                                                                  | 10%         | WB    | 75.9  | 77.8  | 79.1  | 80.6  | 81.0  | 80.6  | 79.9  | 80.1  | 79.9  | 79.7  | 78.4  | 76.1 | (33) |
| (34) |                                                                                  |             | MCDWB | 86.1  | 89.3  | 91.9  | 93.4  | 91.4  | 89.9  | 88.2  | 87.8  | 87.4  | 87.4  | 86.7  | 85.8 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 19.9  | 19.8  | 18.9  | 17.9  | 15.1  | 13.8  | 12.8  | 12.7  | 12.4  | 13.1  | 15.3  | 17.9 | (35) |
| (36) |                                                                                  |             | MCDBR | 20.9  | 20.8  | 21.2  | 20.2  | 17.6  | 15.7  | 14.9  | 14.4  | 14.2  | 14.8  | 16.8  | 18.9 | (36) |
| (37) |                                                                                  |             | MCWBR | 6.8   | 5.3   | 4.8   | 5.0   | 4.8   | 4.4   | 4.5   | 4.6   | 4.6   | 4.8   | 5.4   | 6.2  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 18.4  | 19.0  | 19.2  | 18.7  | 16.1  | 14.4  | 13.9  | 13.6  | 13.3  | 13.7  | 15.2  | 17.1 | (38) |
| (39) |                                                                                  |             | MCWBR | 6.7   | 5.5   | 5.2   | 5.4   | 5.1   | 4.8   | 4.8   | 5.0   | 4.8   | 5.0   | 5.3   | 6.3  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.546 | 0.626 | 0.642 | 0.575 | 0.488 | 0.470 | 0.465 | 0.468 | 0.488 | 0.530 | 0.484 | 0.476 | (40) |      |
| (41) |                                                                                  | taud        | 1.926 | 1.760 | 1.767 | 1.968 | 2.251 | 2.304 | 2.315 | 2.289 | 2.223 | 2.076 | 2.190 | 2.163 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 239   | 226   | 226   | 242   | 259   | 261   | 263   | 265   | 262   | 248   | 256   | 256   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 59    | 73    | 74    | 60    | 44    | 42    | 41    | 43    | 46    | 52    | 45    | 46    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1577  | 1635  | 1661  | 1770  | 1689  | 1601  | 1530  | 1515  | 1455  | 1492  | 1538  | 1529  | (44) |      |
| (45) |                                                                                  | RadStd      | 95    | 74    | 74    | 76    | 89    | 65    | 91    | 94    | 73    | 92    | 98    | 59    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | +0.52 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (21 neighbors) | N/A   | N/A     | N/A    | N/A     | +0.40 | +0.36 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page